

Mapping

BU-ISCIII

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Mapping against reference genome and Variant Calling :

- Mapping vs Alignment
- What is mapping?
- How to choose a NGS mapper.
- SAM/BAM format
- Duplicate filter

Alignment

Definition:

Arrange two or more nucleotide or aminoacid sequences to identify regions of similarity that may be a consequence of functional, structural, or evolutionary relationships.

```
AAB24882      TYHMCQFHC RYVNNHSGEKL YECNERSKAFSCPSHLQCHKRRQIG EKT HEHNQCGKAFPT
AAB24881      -----YECNQCGKAFAQHSSLKCHYRTHIG EKPYECNQCGKAFSK

AAB24882      PSHLQYHERTHTG EKPYECHQCGQAFKKCSLLQRHKRTHTG EKPYE -CNQCGKAFAQ-
AAB24881      HSHLQCHKRTHTG EKPYECNQCGKAFSQHGLLQRHKRTHTG EKPYMNVINMVKPLHNS
```

Multiple alignment (MSA)

Definition:

A multiple alignment is a collection of three or more sequences partial or completely aligned.

File Edit Colour Sort Picked: Column 50: seq_cons/0-0 c = 48 (1 match)			
(16x225) -----10-----20-----30-----40-----50-----60-----70			
ALSE_ECOLI	2	202	KISPSLMCM DLLKFKEQIEFIDS.HADYFHIDIMDGHFVPNLTLSEFFVSQVKKL.....A
RPE_YEAST	5	214	IIAPSILASDFANLGCECHKVINAGADWLHIDVMDGHFVPNITLGGPIVTSLLRSVPRPGDASNTEKKP
O14105	5	204	KIAPSLLAGDFANLEKEVGRMLKYGSOWLHVDVMDAQFVPNITIGPIVVKAMRNHYT.....KE
RPE_SYNY3	5	207	VVAPSILSADFSRLGEEIKAVDEAGADWIHVDVMDGRFVPNITIGPLIVDAIRPL.....TI
RPE_SOLTU	58	260	IVSPSILSANFSKLGEQVKAIEQAGCDWIHVDVMDGRFVPNITIGPLVVDLSLRPI.....TI
RPE_BACSU	3	204	KVAPSILSADFAALGNEIKDVEKGGADCIHIDVMDGHFVPNITIGPLIVEAVRPV.....TI
RPE_HAEIN	5	206	LIAPSILSADLARLGDDVQNVNLNAGADVIHFDVMDNHYVPNLTFGPAVCQALRDYG.....I
RPE_ECOLI	5	206	LIAPSILSADFARLGEDTAKALAAGADVHFDVMDNHYVPNLTFGPMVLKSLRNYG.....I
RPEC_ALCEU	17	221	RLAPSILSADFARLGEEVCAIEAGGADLVHFDVMDNHYVPNLTFGLVCEAIRPL.....VS
RPE_RHORU	6	204	RIAPSLLSADFAISRPRCPSDGRTGADILHFDVMDNHYVPNLTVGLVCAALRPH.....TS
RPE_MYCTU	9	207	LIAPSILAADFARLADEAAAVN..GADWLHVDVMDGHFVPNLTFGLPVVESLLAVTD.....IP
RPE_HELPY	2	200	KVAPSLLSADFMHLAKEIESVSN..DFLHVDVMDGHYVPNLTMGFVWLENVTQM.....S
RPE_METJA	3	201	KIGASILSADFGHLREEIKKAEAGVDFHVDVMDGHFVPNISMGIGIAKHVKKL.....TE
SGCE_ECOLI	2	198	ILHPSILASANPLHYGRELTALDNLDGSLHLIDIEDSSFINNITFGMKTVQAVARQ.....TF
RPE_MYCPN	9	203	EIAFSLPLLLHQFDRKLLLEQFFADGLRLIHYDVMD.HFVDNTVFQGEHLDELQIG.....
RPE_MYCGE	15	198	RFDKSLLSEYFQDGLRLIHYDVMD.QFVHNTAFKGEYLDDELKTIG.....

Mapping definition

Definición:

Place a sequence inside a larger sequence. For example, determine the position of a read inside a reference genome.

Referencia/ genoma

```
...GTGGGCCGGCAATTCGATATCGCGCATATATTCGGGCGCATGCTTAGC...
```

Lecturas:

```
GCAATTCGATAT
GCGCATATATTT
TGGGCCGGCAAT
CGCATGCTTAGC
ATTCGATATCGC
GCCGGCAATTCG
```

Mapeo

```
...GTGGGCCGGCAATTCGATATCGCGCATATATTCGGGCGCATGCTTAGC...
```

```
      GCAATTCGATAT          CGCATGCTTAGC
```

```
TGGGCCGGCAAT      GCGCATATATTT
```

```
      ATTCGATATCGC
```

```
GCCGGCAATTCG
```

Alignment vs mapping

Mapping:

- A mapping is regarded to be correct if it overlaps the true region.
- Each read maps independently
- From thousand to millions of sequences.

Multiple alignment:

- An alignment is regarded to be correct only if each base is placed correctly.
- Minimizes differences among sequences
- From tens to hundred of sequences.

Consideration:

An algorithm can be good at mapping but may not be good aligning. This is because the true alignment minimizes differences between reads, but the read mapper only sees the reference.

So in summary...

CTGACCTCATGTGATCCACCCGCCTTGGCC

Find best match for the read
in a reference sequence

TGATCCAC

Challenges

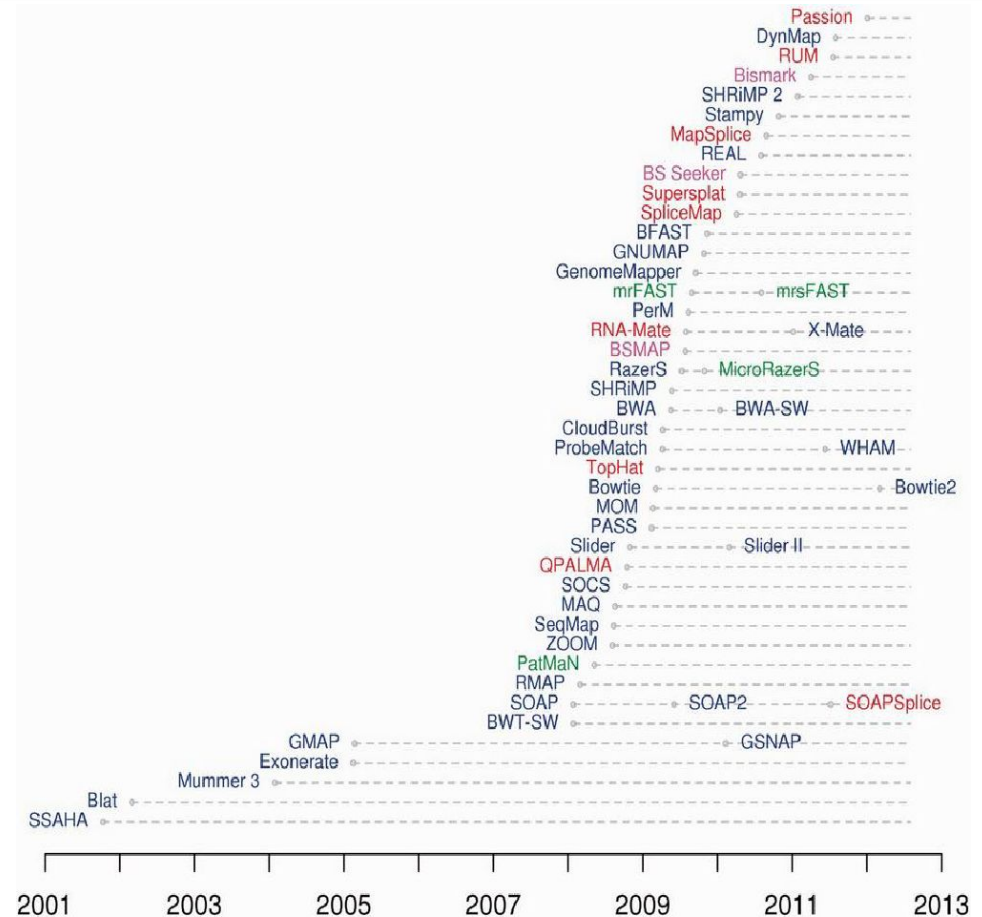
- Errors in reads
- Errors in libraries
- Repetitive regions (repeats, homologous regions)
- Homopolymers
- Individual polymorphisms

Pierre Lechat. Variants Calling lecture. Pasteur.fr

What mapper should I use?

Mappers:

- Más de 60 mappers available.
- Lots of papers reviewing its performamnce.



What mapper should I use?

Things to consider:

- Computational resources vs sensibility
- Platform and type of experiment (Illumina/454/etc, paired-end, DNA/RNA/etc)
- Variation (indels allowance, mismatch number, etc.)
- Repetitions (all regions, best match, random, user

Important:

Default options don't have to be the best:

“... there is no tool that outperforms all of the others in all the tests. Therefore, the end user should clearly specify his needs in order to choose the tool that provides the best results.” - Hatem et al *BMC Bioinformatics* 2013, **14**:184

End-to-end vs local alignment

End-to-end

Read: GACTGGGCGATCTCGACTTCG

Reference: GACTGCGATCTCGACATCG

Alignment:

Read: GACTGGGCGATCTCGACTTCG

|||| | ||||| |||

Reference: GACTG--CGATCTCGACATCG

Local

Read: ACGGTTGCGTTAATCCGCCACG

Reference: TAACTTGCGTTAAATCCGCCTGG

Alignment:

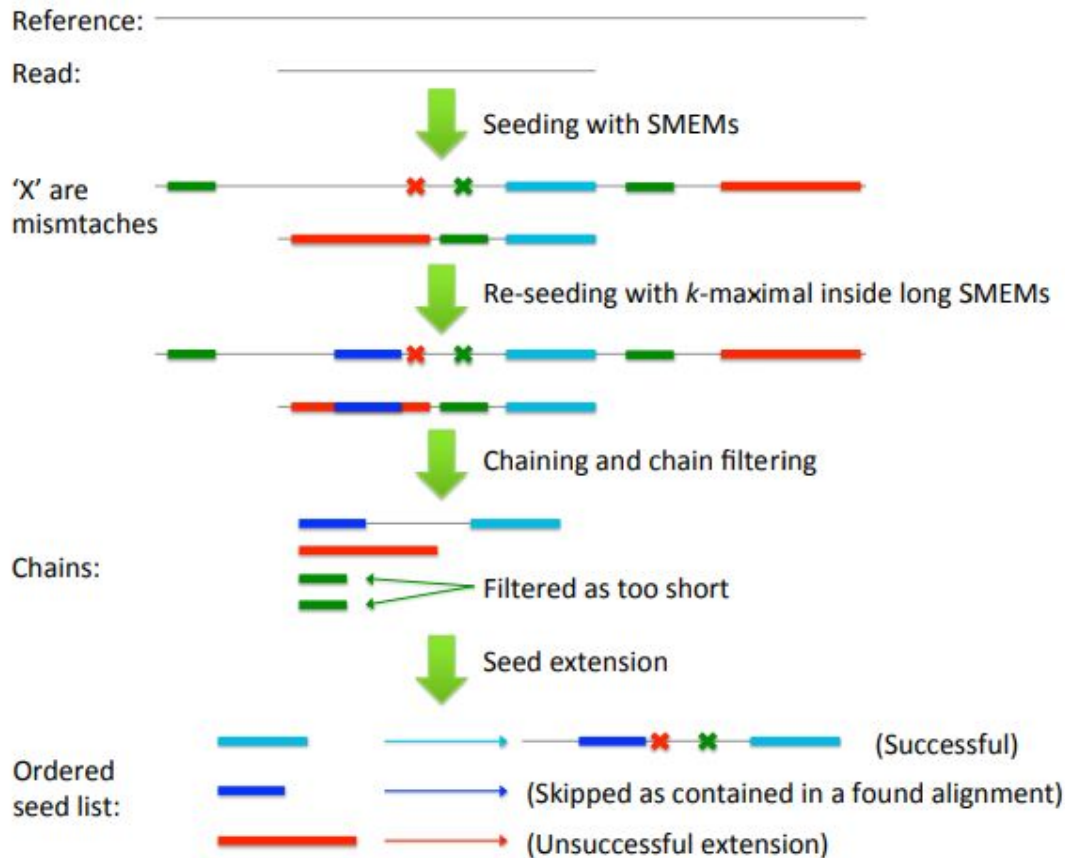
Read: ACGGTTGCGTTAA-TCCGCCACG

||||| |||||

Reference: TAACTTGCGTTAAATCCGCCTGG

[Bowtie2 manual.](#)

BWA MEM



SMEM strategy

- Maximal exact match (MEM): an exact match that cannot be extended further in either direction
- Super-maximal exact match (SMEM): a MEM that is not contained in any other MEMs on the query coordinate (Li, 2012). At any query position, the longest exact match covering the position must be a SMEM.

Seed-and-extend algorithm

Local alignment

Hen LI. Aligning sequence reads, clone sequences and assembly con*gs with BWA-MEM. Poster. Broad Institute.

BOWTIE2

End-to-end alignment by default.

Three reporting modes:

- Best alignment
- K alignments
- All alignments

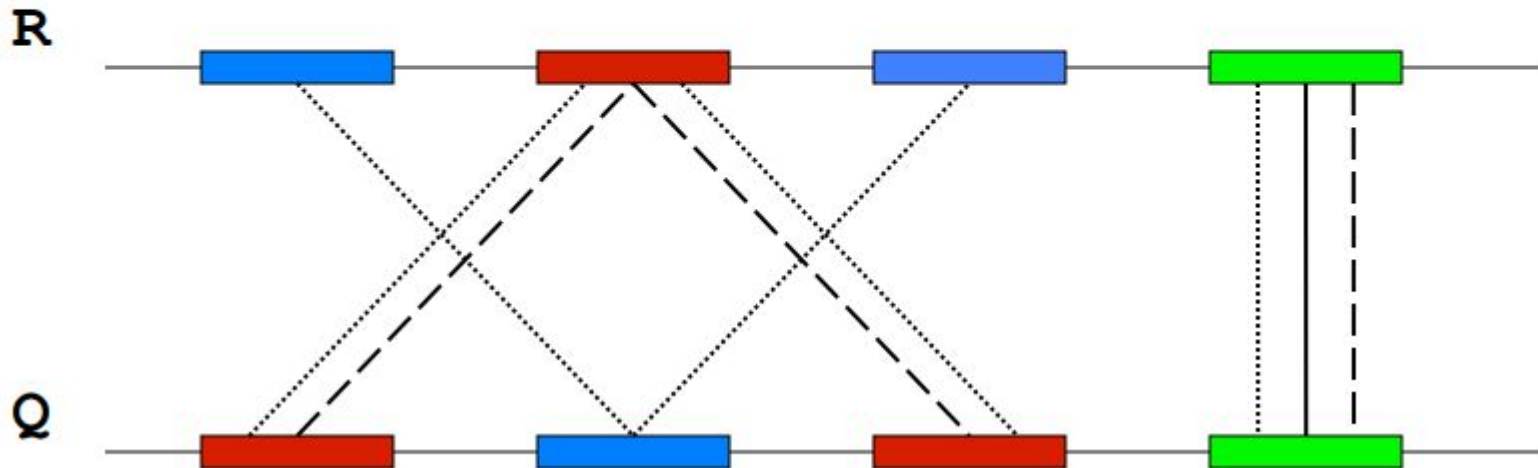
Lots of customizable parameters that change its performance.

Example whole genome aligner: MUMMER

MUM : maximal unique match

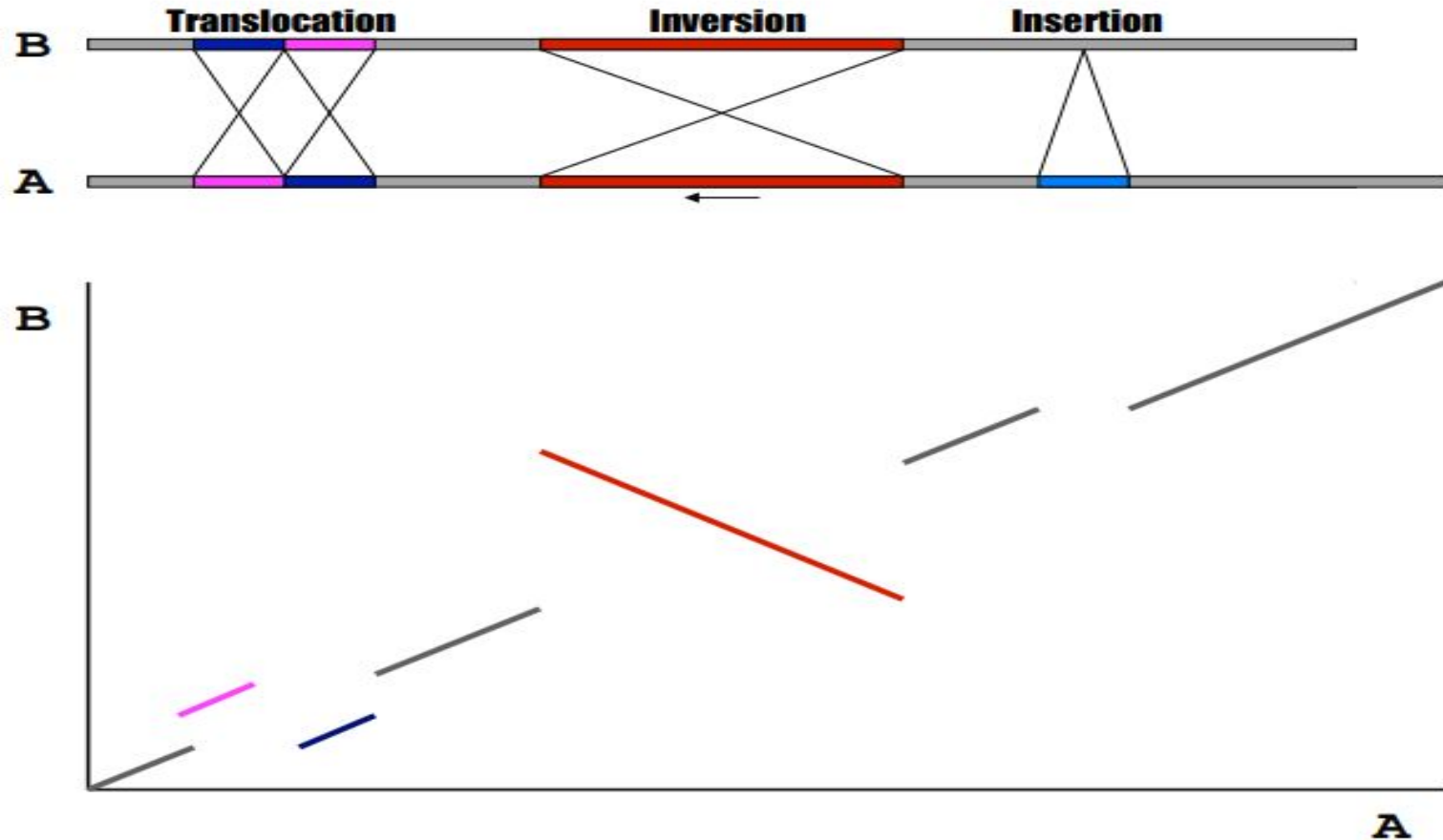
MAM : maximal almost-unique match

MEM : maximal exact match



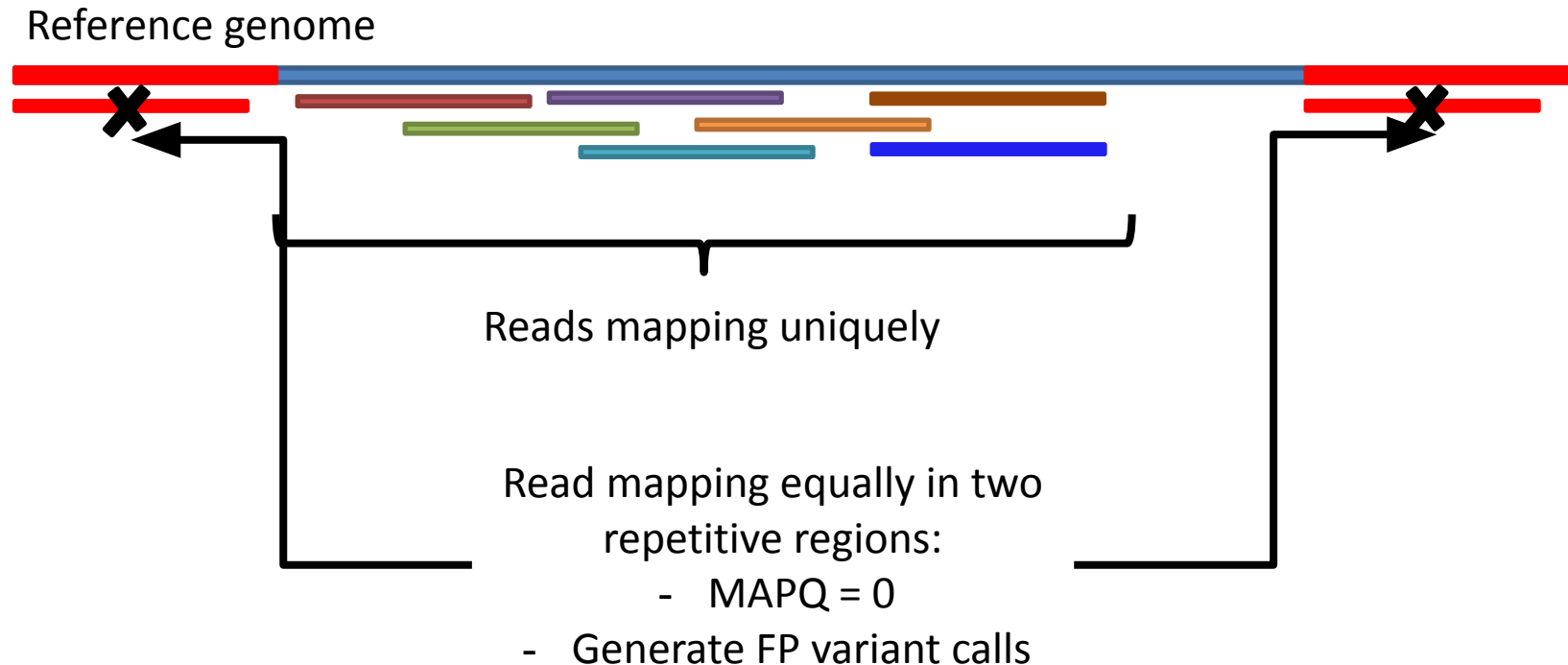
Adam M. Phillippy. Whole Genome Alignment with MUMmer. Presentation.

Example whole genome aligner: MUMMER



Adam M. Phillippy. Whole Genome Alignment with MUMmer. Lecture.

Which aligner should I use for aligning reads against a complete genome for variant calling?



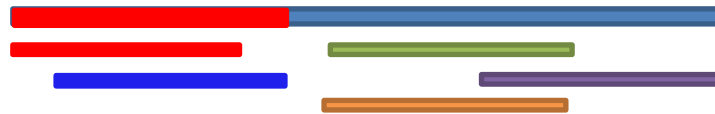
Which aligner should I use for aligning reads against a resistance gene database for determining with resistance genes I have in my sample?


Homologus/repetitive region

Reads mapping to the repetitive/homologus region map against all alleles.

We allow one read to map to **several locations**.

Resistance gene - Allele 1



Reads mapping uniquely only map in Allele 1.
Which is the one more **covered**

Resistance gene - Allele 2



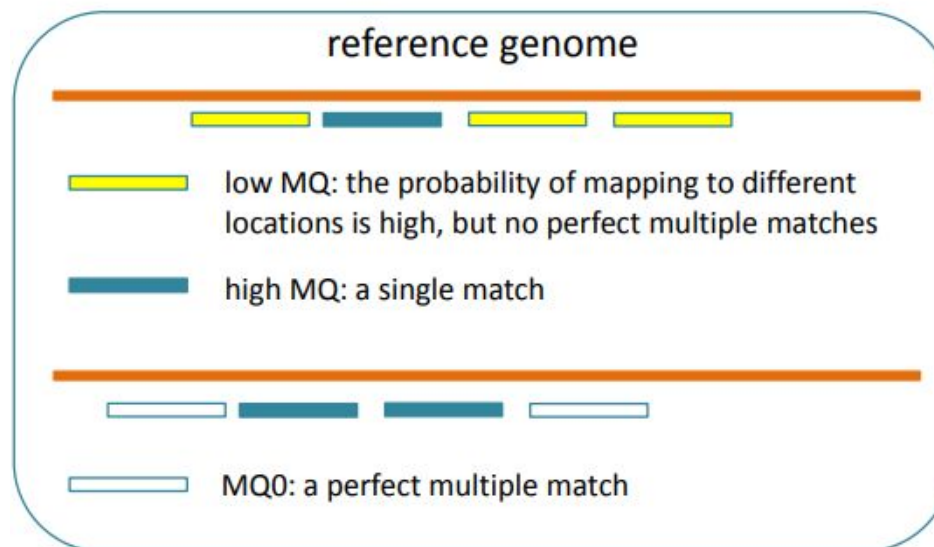
Resistance gene - Allele 3



MAPQ

- What if there are several possible places to align your sequencing read? This may be due to:
 - Repeated elements in the genome
 - Low complexity sequences
 - Reference errors and gaps

MQ is a phredScore of the quality of the alignment



MAPQ

MAPQ is NOT comparable among mappers.

BWA:

- MAPQ represents the probability of the read to be mapped correctly.
- MAPQ = 0 identifies unmapped reads and...

Reads mapping to different locations!

BOWTIE2:

- MAPQ represents the “uniqueness” of the read. A MAPQ < 10 indicates that there is at least a 1 in 10 chance that the read truly originated elsewhere
- MAPQ = 0 identifies unmapped reads

SAM format

Definición:

It's a specification that defines a generic format for storing nucleotide alignments. It describes a query alignment against a reference genome.

```
@HD VN:1.5 S0:coordinate
@SQ SN:ref LN:45
r001 99 ref 7 30 8M2I4M1D3M = 37 39 TTAGATAAAGGATACTG *
r002 0 ref 9 30 3S6M1P1I4M * 0 0 AAAAGATAAGGATA *
r003 0 ref 9 30 5S6M * 0 0 GCCTAAGCTAA * SA:Z:ref,29,-,6H5M,17,0;
r004 0 ref 16 30 6M14N5M * 0 0 ATAGCTTCAGC *
r003 2064 ref 29 17 6H5M * 0 0 TAGGC * SA:Z:ref,9,+,5S6M,30,1;
r001 147 ref 37 30 9M = 7 -39 CAGCGGCAT * NM:i:1
```

SAM format

Col	Field	Type	Regexp/Range	Brief description
1	QNAME	String	[!-?A-~]{1,255}	Query template NAME
2	FLAG	Int	[0,2 ¹⁶ -1]	bitwise FLAG
3	RNAME	String	* [!-()+-<>-~] [!-~]*	Reference sequence NAME
4	POS	Int	[0,2 ³¹ -1]	1-based leftmost mapping POSition
5	MAPQ	Int	[0,2 ⁸ -1]	MAPping Quality
6	CIGAR	String	* ([0-9]+[MIDNSHPX=])+	CIGAR string
7	RNEXT	String	* = [!-()+-<>-~] [!-~]*	Ref. name of the mate/next read
8	PNEXT	Int	[0,2 ³¹ -1]	Position of the mate/next read
9	TLEN	Int	[-2 ³¹ +1,2 ³¹ -1]	observed Template LENgth
10	SEQ	String	* [A-Za-z=.]+	segment SEQUENCE
11	QUAL	String	[!-~]+	ASCII of Phred-scaled base QUALity+33

@HD VN:1.5 S0:coordinate

@SQ SN:ref LN:45

```
r001 99 ref 7 30 8M2I4M1D3M = 37 39 TTAGATAAAGGATACTG *
r002 0 ref 9 30 3S6M1P1I4M * 0 0 AAAAGATAAGGATA *
r003 0 ref 9 30 5S6M * 0 0 GCCTAAGCTAA * SA:Z:ref,29,-,6H5M,17,0;
r004 0 ref 16 30 6M14N5M * 0 0 ATAGCTTCAGC *
r003 2064 ref 29 17 6H5M * 0 0 TAGGC * SA:Z:ref,9,+,5S6M,30,1;
r001 147 ref 37 30 9M = 7 -39 CAGCGGCAT * NM:i:1
```

SAM format: flags

Bit	Description
0x1	template having multiple segments in sequencing
0x2	each segment properly aligned according to the aligner
0x4	segment unmapped
0x8	next segment in the template unmapped
0x10	SEQ being reverse complemented
0x20	SEQ of the next segment in the template being reversed
0x40	the first segment in the template
0x80	the last segment in the template
0x100	secondary alignment
0x200	not passing quality controls
0x400	PCR or optical duplicate
0x800	supplementary alignment

<https://broadinstitute.github.io/picard/explain-flags.html>

Flag explanation example 1

SAM Flag:

Toggle first in pair / second in pair

Find SAM flag by property:

To find out what the SAM flag value would be for a given combination of properties, tick the boxes for those that you'd like to include. The flag value will be shown in the SAM Flag field above.

- ☒ read paired
- ☒ read mapped in proper pair
- ☐ read unmapped
- ☐ mate unmapped
- ☐ read reverse strand
- ☒ mate reverse strand
- ☒ first in pair
- ☐ second in pair
- ☐ not primary alignment
- ☐ read fails platform/vendor quality checks
- ☐ read is PCR or optical duplicate
- ☐ supplementary alignment

Summary:

- read paired (0x1)
- read mapped in proper pair (0x2)
- mate reverse strand (0x20)
- first in pair (0x40)

Flag explanation example 2

SAM Flag:

Toggle first in pair / second in pair

Find SAM flag by property:
To find out what the SAM flag value would be for a given combination of properties, tick the boxes for those that you'd like to include. The flag value will be shown in the SAM Flag field above.

- ☒ read paired
- ☒ read mapped in proper pair
- ☐ read unmapped
- ☐ mate unmapped
- ☒ read reverse strand
- ☐ mate reverse strand
- ☐ first in pair
- ☒ second in pair
- ☐ not primary alignment
- ☐ read fails platform/vendor quality checks
- ☐ read is PCR or optical duplicate
- ☐ supplementary alignment

Summary:
read paired (0x1)
read mapped in proper pair (0x2)
read reverse strand (0x10)
second in pair (0x80)

SAM format: CIGAR string

Op	BAM	Description
M	0	alignment match (can be a sequence match or mismatch)
I	1	insertion to the reference
D	2	deletion from the reference
N	3	skipped region from the reference
S	4	soft clipping (clipped sequences present in SEQ)
H	5	hard clipping (clipped sequences NOT present in SEQ)
P	6	padding (silent deletion from padded reference)
=	7	sequence match
X	8	sequence mismatch

SAM vs BAM format

- SAM and BAM format are exactly the same.
 - SAM is a tabular plain text file.
 - BAM is its binary format. Binary meaning is in a compress format not human readable.
 - We **MUST** always use BAM format because it is optimized for computer-reading

AND

BECAUSE IT SAVES A LOT OF DISK SPACE!!

Typical bam and sam format files weights from a *S. grumpensis*

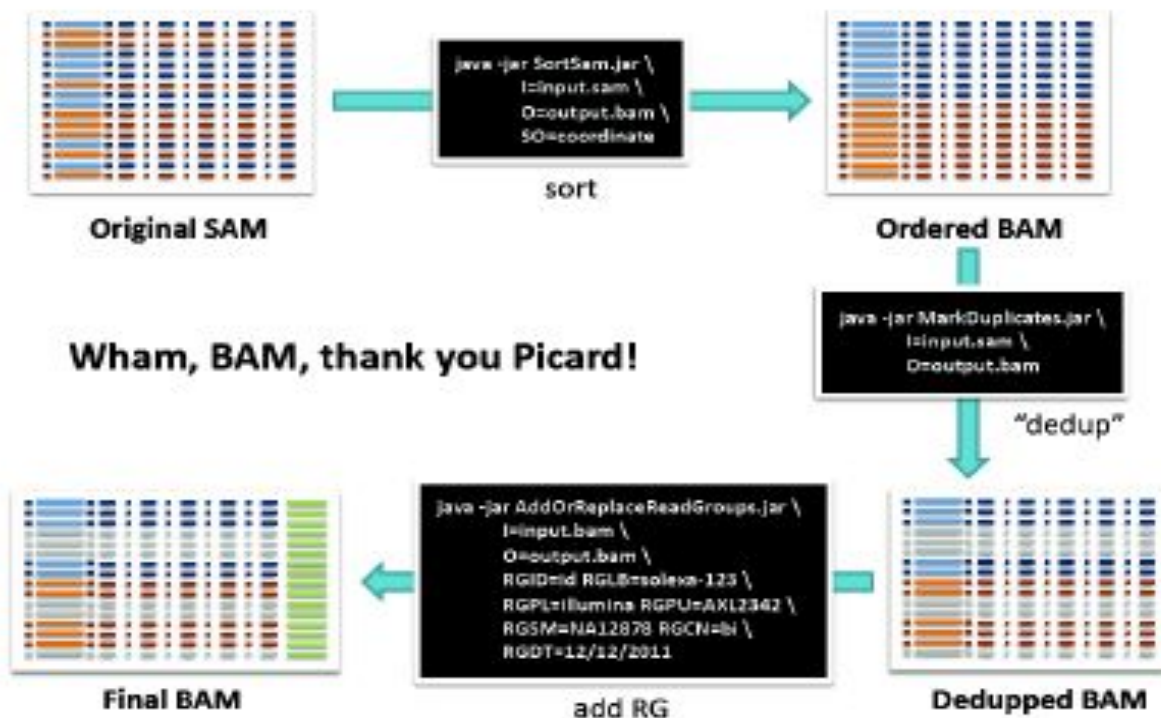
SAM format file: 3.6 GB

BAM format file: 689 M

Duplicate filter

- Duplicates are non-independent measurements of a sequence
 - Sampled from the exact same template of DNA
 - Violates assumptions of variant calling
- Errors in sample/library prep will get propagated to all the duplicates
- Just pick the “best” copy – mitigates the effects of errors
- **Definition:** sequences starting and finishing in the exact same coordinates. Both pairs if paired-end.

Duplicate filter



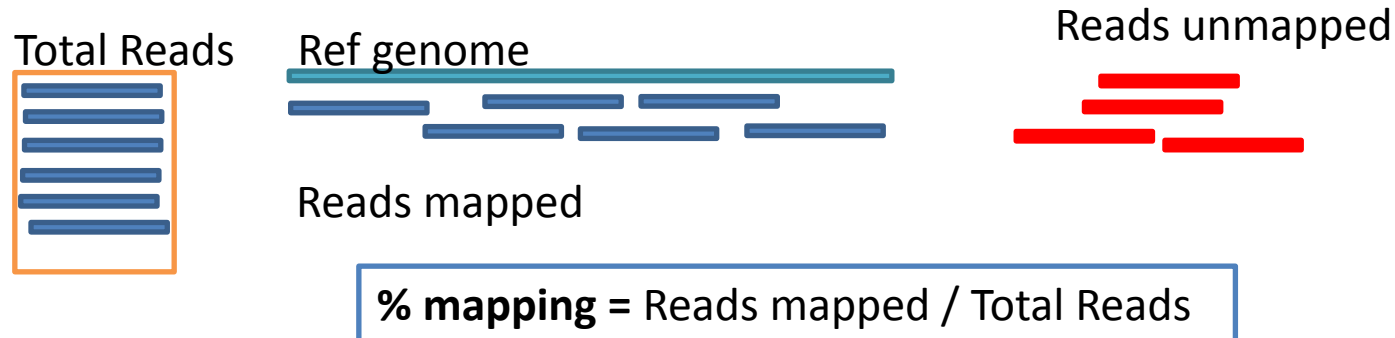
Mapping statistics

- % mapped: $\text{reads mapped} / \text{total reads}$
- % unmapped: $\text{reads unmapped} / \text{total reads}$
- % duplicates: $\text{reads belonging to same template} / \text{total reads}$
- Mean depth of coverage
- Coverage: % genome with at least one read mapped.

Mapping quality control

- **% mapping:** number of reads mapping againsts reference genome.

Picard
Samtools

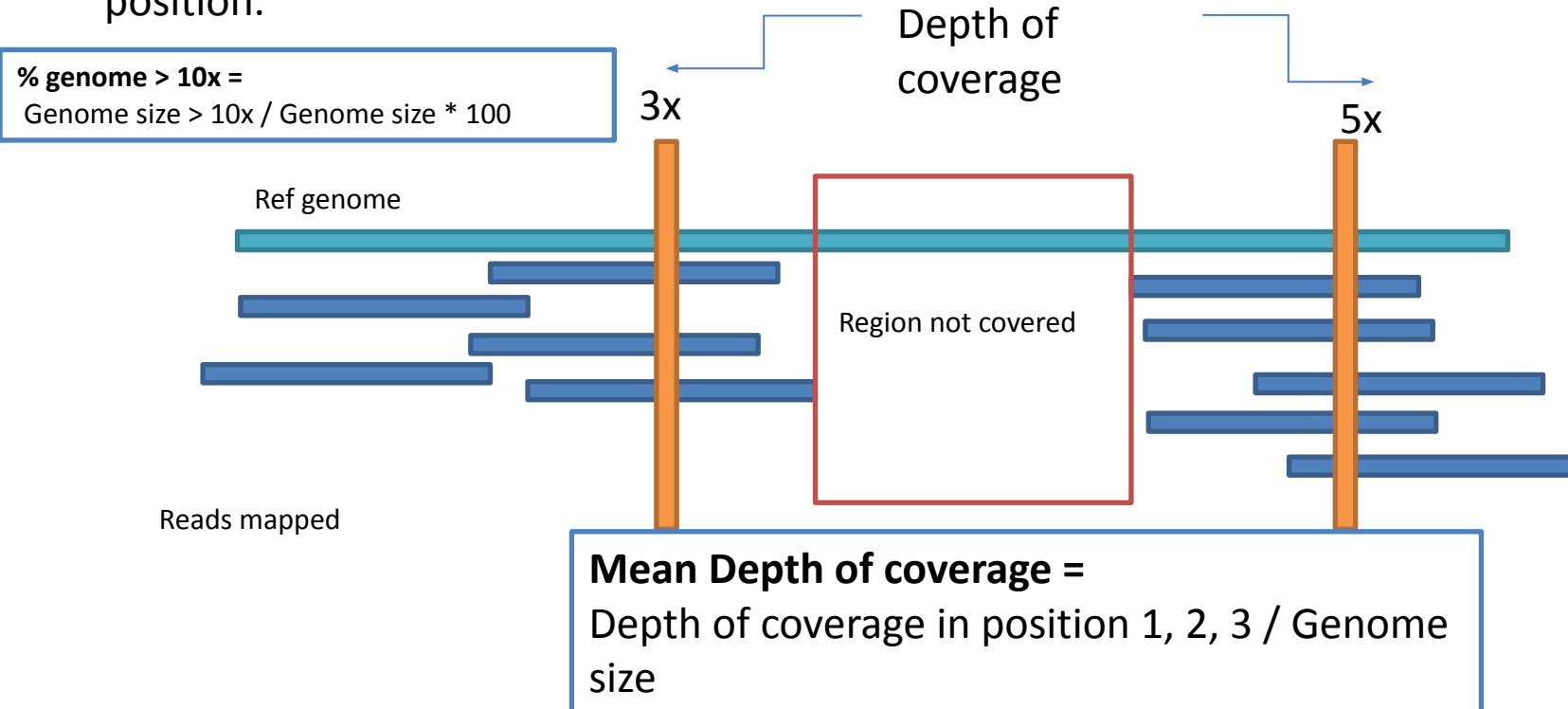


Mandatory parameter for microbial genomics!! It indicates us how many reads we have from our organism of interest. In human genomics this is almost always 99.99% unless something terrible happens. Not here!!!

Mapping quality control

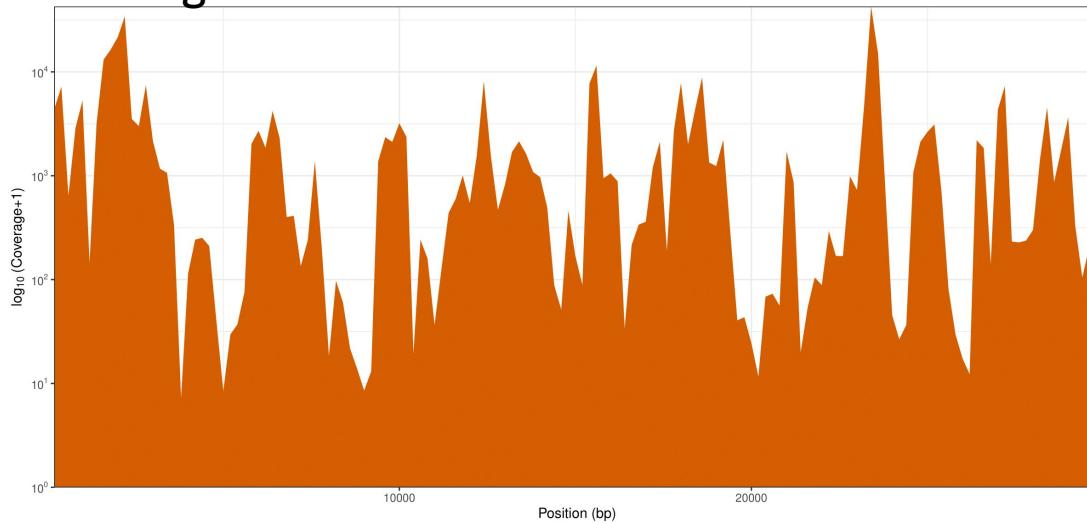
- **% genome > 10x:** percentage of genome covered with more than 10 reads.
- **Mean Depth of coverage:** mean of reads covering a genome position.

Picard
Samtools

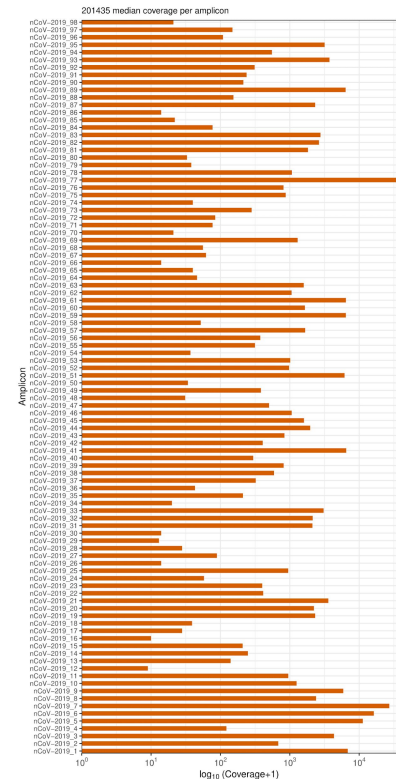


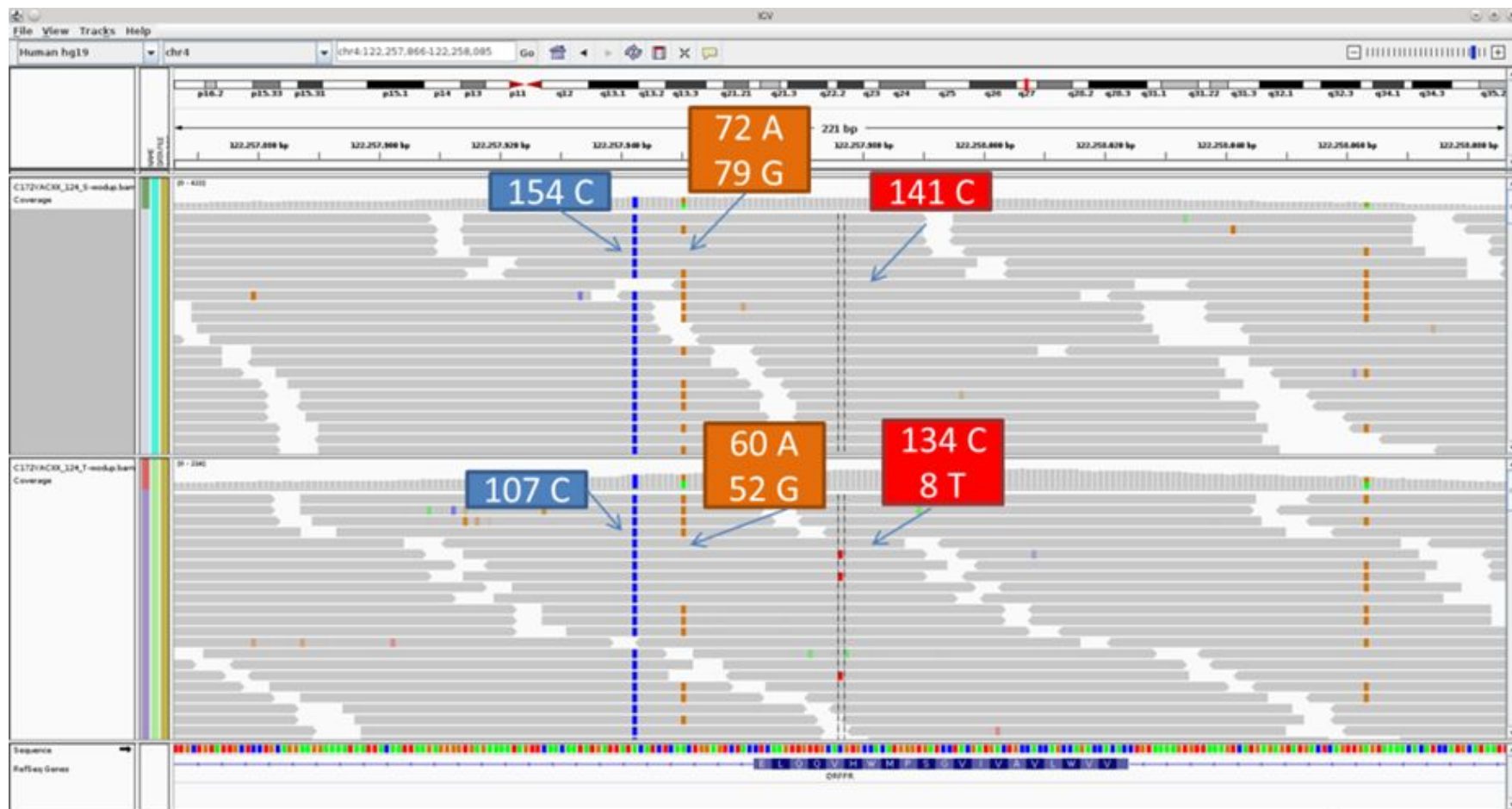
Amplicon QC results

Genome coverage



Amplicon coverage





Thanks for your attention!
